Recitation 9 (Section 005)

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November 4, 2026

University of Minnesota

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Today's Agenda

- Anouncements
- Review Questions
- ICA

Announcements: Midterm 2

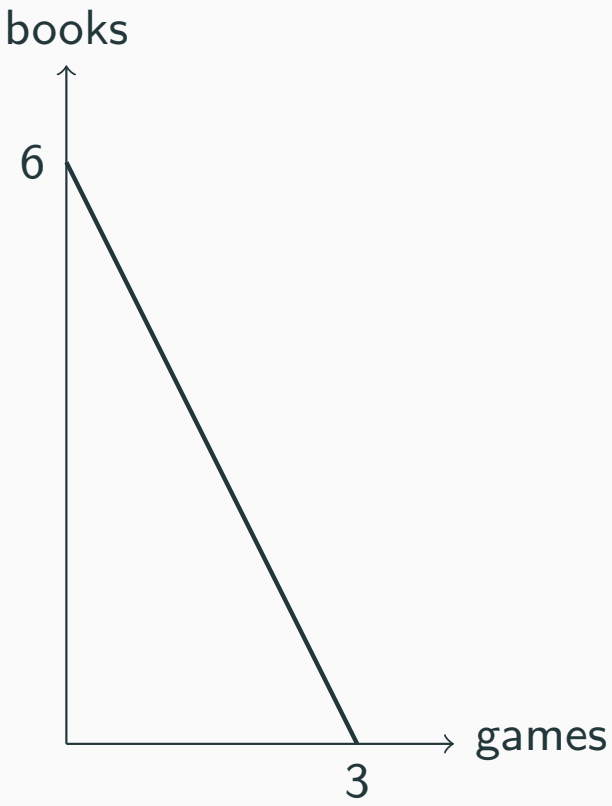
- On **Monday, November 16th**.
- Please register for the makeup by Nov 11th if you need to.
- Exam review session times will be announced later.
- Please tell me about ways you may or may not have felt blindsided by midterm 1 so that I can change up my content before the next one!

Quick Concept Review: Consumer Choice

Before today's ICA, let's warm up on budget lines and indifference curves:

- Reading a budget line (opportunity cost)
- What indifference curves represent
- Using both *together* to find the optimum
- Income changes: normal vs. inferior goods
- Price changes: substitution vs. income effect

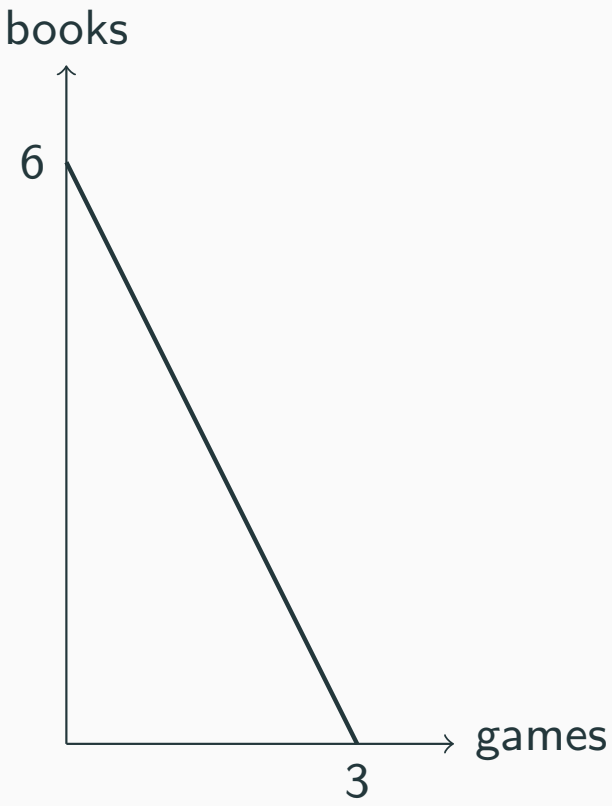
Warm-Up 1: Reading a Budget Line



Maya has \$60 to spend on video games (\$20 each) and books (\$10 each).

What is the slope of her budget line, and what does it represent?

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What is the slope of her budget line, and what does it represent?

Answer: Slope $= -2$. Giving up 1 game frees up \$20, which buys 2 books. The slope is always $-P_{games}/P_{books} = -20/10 = -2$: the opportunity cost of a game, in books.

Warm-Up 2: True or False?

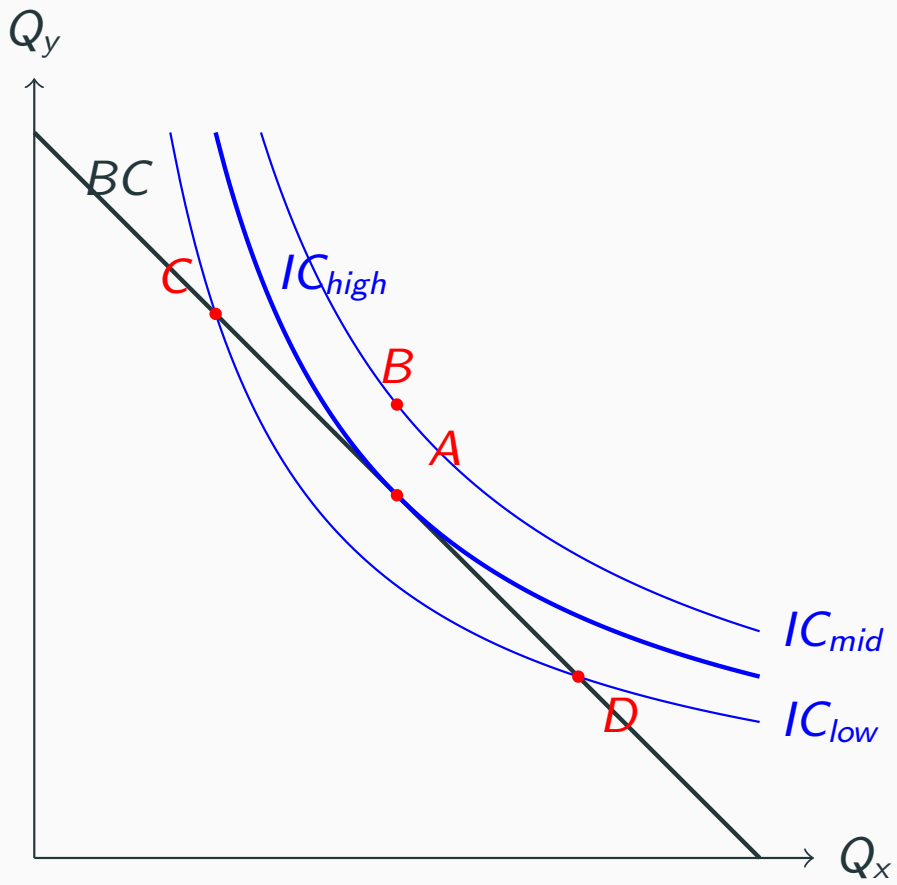
1. A consumer is always better off on a higher indifference curve.
2. Two indifference curves can cross, as long as they represent different goods.
3. Indifference curves are bowed *outward* (convex to the origin), not inward.
4. The slope of an indifference curve is the marginal rate of substitution (MRS).

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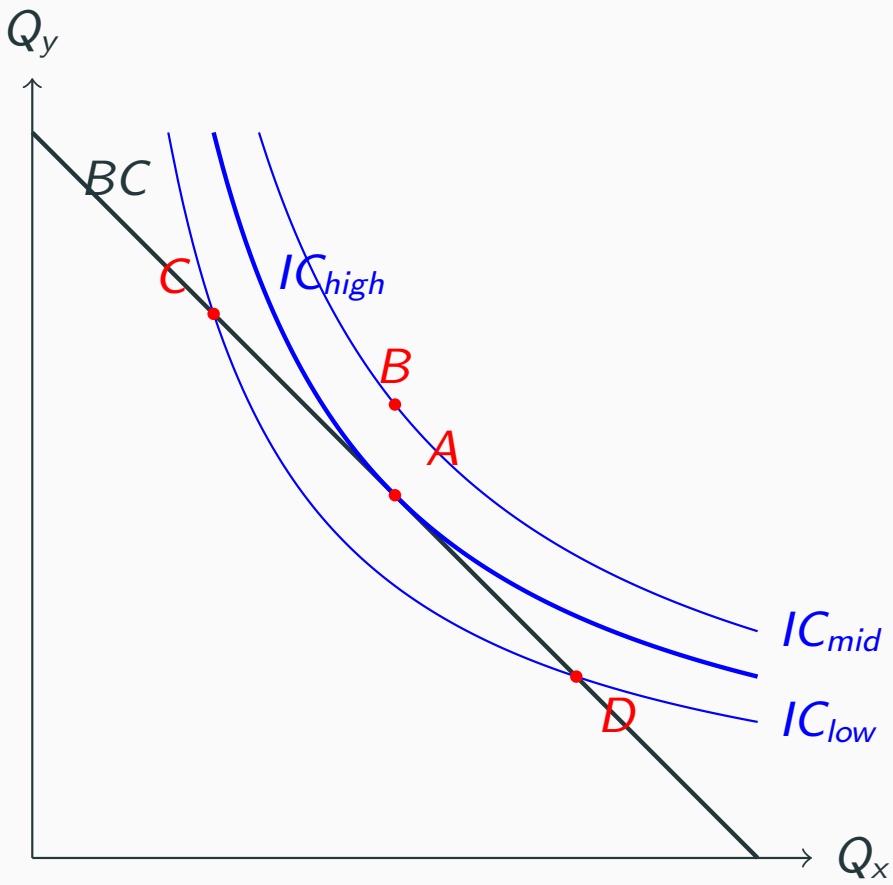
Answer: (1) **True**. (2) **False** — if two curves crossed, the crossing bundle would have to give two different utility levels, a contradiction. (3) **False** — they are bowed *inward* (toward the origin). (4) **True**.

Warm-Up 3: Finding the Optimum



Points A, B, C, D: which one is the consumer's optimal choice? What's wrong with the other three?

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Points A, B, C, D: which one is the consumer's optimal choice? What's wrong with the other three?

Answer: A is optimal: it's on the budget line *and* tangent to the highest reachable curve. B is preferred to A but unaffordable. C and D are affordable but on a lower indifference curve than A — the consumer can do strictly better by moving to A.

Warm-Up 4: Why Are Indifference Curves Bowed In?

You have **10 pairs of socks** and only **1 pair of shoes**.

Are you more willing to give up a pair of socks, or your one pair of shoes, to get one more unit of the other good?

Now suppose it's reversed: 10 pairs of shoes, 1 pair of socks. Does your answer flip?

Warm-Up 4: Why Are Indifference Curves Bowed In?

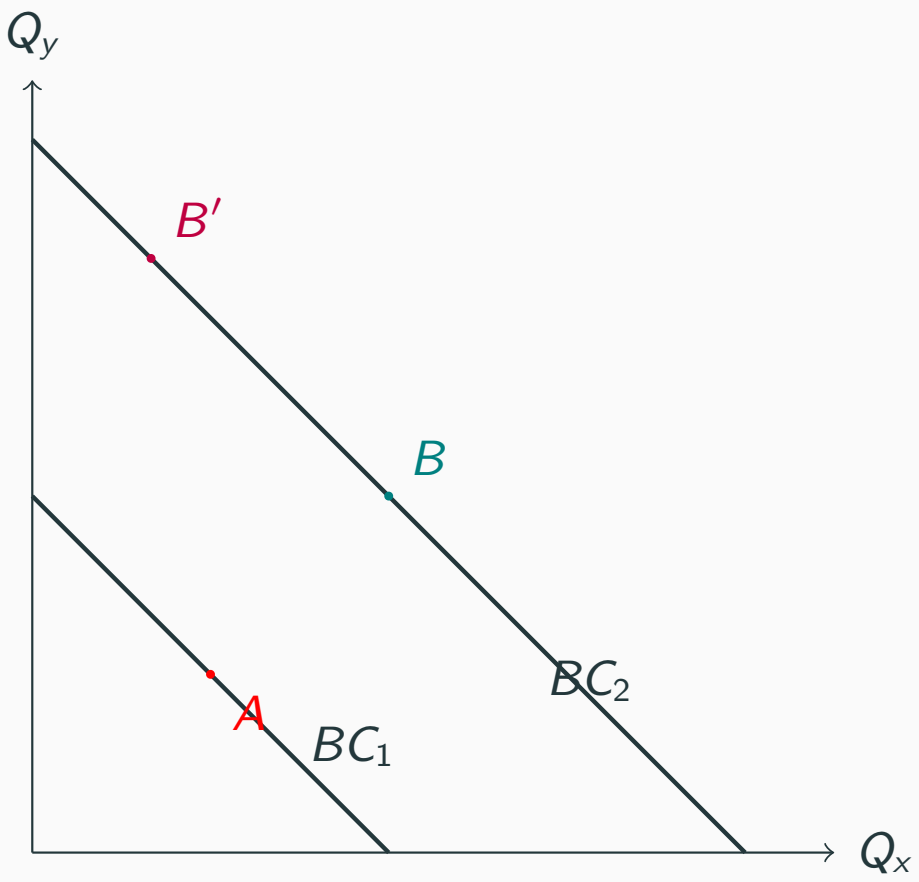
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Now suppose it's reversed: 10 pairs of shoes, 1 pair of socks. Does your answer flip?

Answer: You'll happily give up socks (you have plenty) but not your only pair of shoes. With the ratio reversed, your willingness flips too. This is **diminishing MRS**: you're more willing to trade away whatever good you have in abundance — which is exactly why indifference curves get flatter as you move right (more x, less y).

Warm-Up 5: Normal or Inferior?



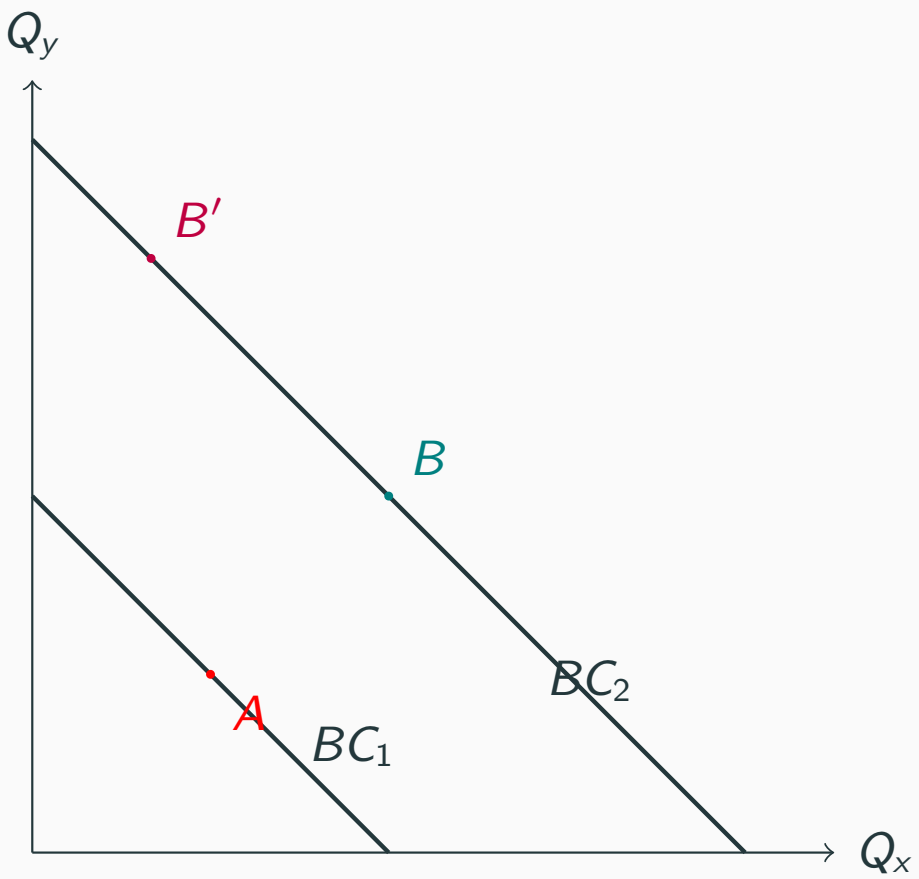
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Income rises. Original optimum is A.

If the new optimum is B: is x normal or inferior?

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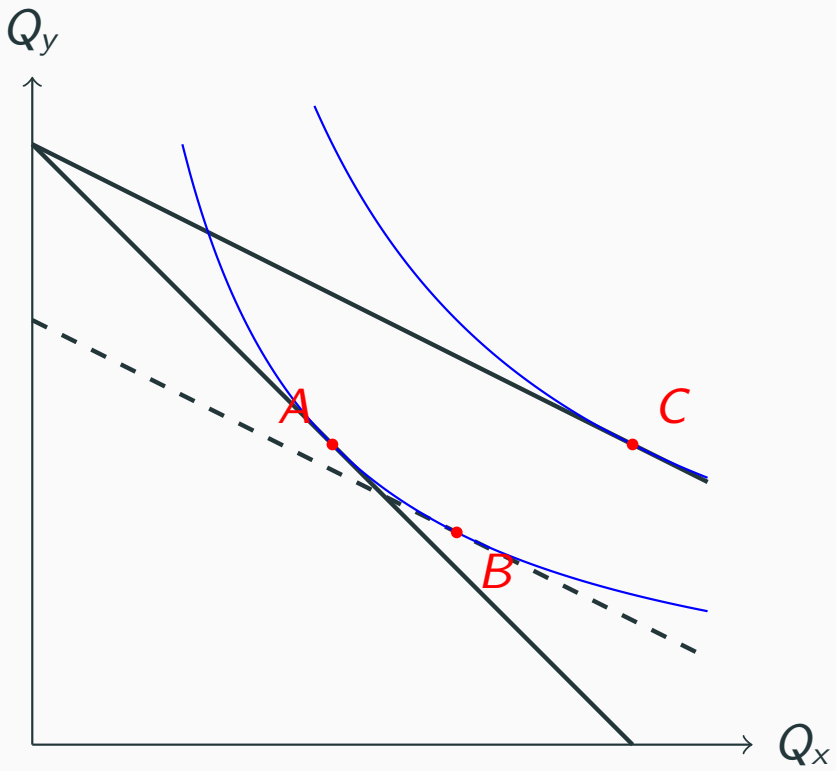
Income rises. Original optimum is A.

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Answer: At B: x rises (3→6) — **normal**. At B': x falls (3→2) despite the income rise — **inferior**.

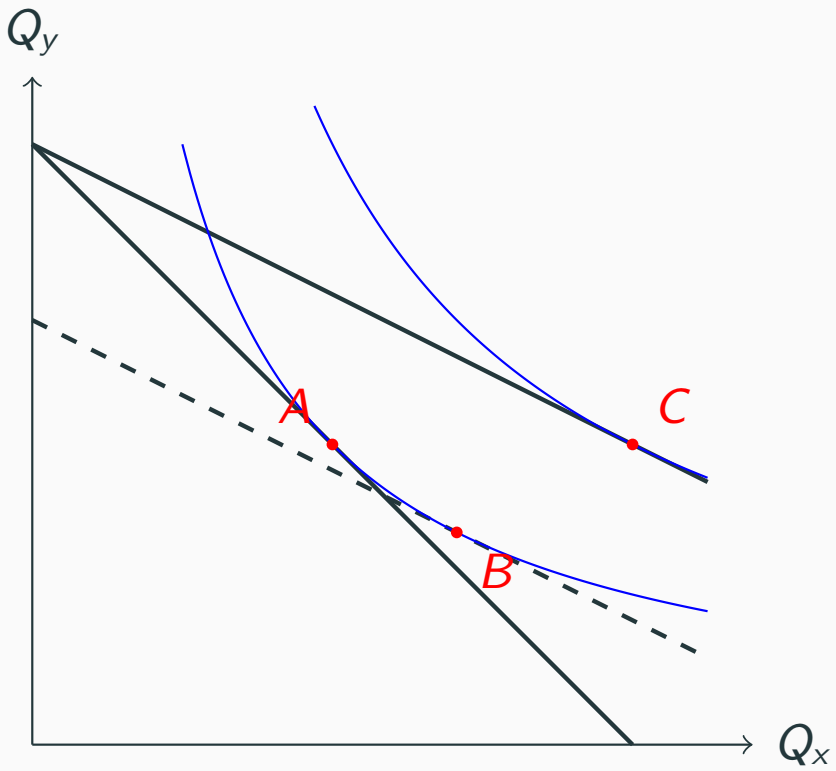
Warm-Up 6: Substitution vs. Income Effect



The price of x falls. A is the original optimum, C is the new optimum. B is on the *original* indifference curve, using the *new* price ratio (the dashed line).

Which move, $A \to B$ or $B \to C$, is the substitution effect? Which is the income effect? Is x normal or inferior here?

Warm-Up 6: Substitution vs. Income Effect



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Which move, $A \to B$ or $B \to C$, is the substitution effect? Which is the income effect? Is x normal or inferior here?

Answer: $A \to B$ is the **substitution effect** (same utility, new relative price). $B \to C$ is the **income effect** (purchasing power rises). Here x rises from B to C too, so x is **normal**.

Warm-Up 7: International Trade, Revisited

Before trade, the domestic price of steel in Isoland is \$600/ton. The world price is \$450/ton.

1. Once trade opens, will Isoland import or export steel?
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3. What happens to the domestic price of steel?

Answer: Domestic price $>$ world price, so Isoland **imports** steel. The rest of the world has the comparative advantage (lower opportunity cost). The domestic price **falls** to \$450.

Warm-Up 8: Opening Up to Trade (Numbers)

Domestic demand for widgets is $Q^D = 100 - P$ and domestic supply is $Q^S = P - 20$.

1. Find the autarky price and quantity. Compute CS, PS, and total surplus.
2. The world price is $P^W = 40$. Find Q^D, Q^S, and imports.
3. Compute CS, PS, and total surplus with free trade. What are the gains from trade?

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Answer: (1) $P = 60$, $Q = 40$. CS $= \frac{1}{2}(40)(40) = 800$, PS $= \frac{1}{2}(40)(40) = 800$, TS $= 1600$.
(2) $Q^D = 60$, $Q^S = 20$, imports $= 40$.
(3) CS $= \frac{1}{2}(60)(60) = 1800$, PS $= \frac{1}{2}(20)(20) = 200$, TS $= 2000$. Gains from trade $= 400 = \frac{1}{2}(20)(40)$.

Open up Canvas and navigate to **“In-Class Activity 7 (Discussions)”**!

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